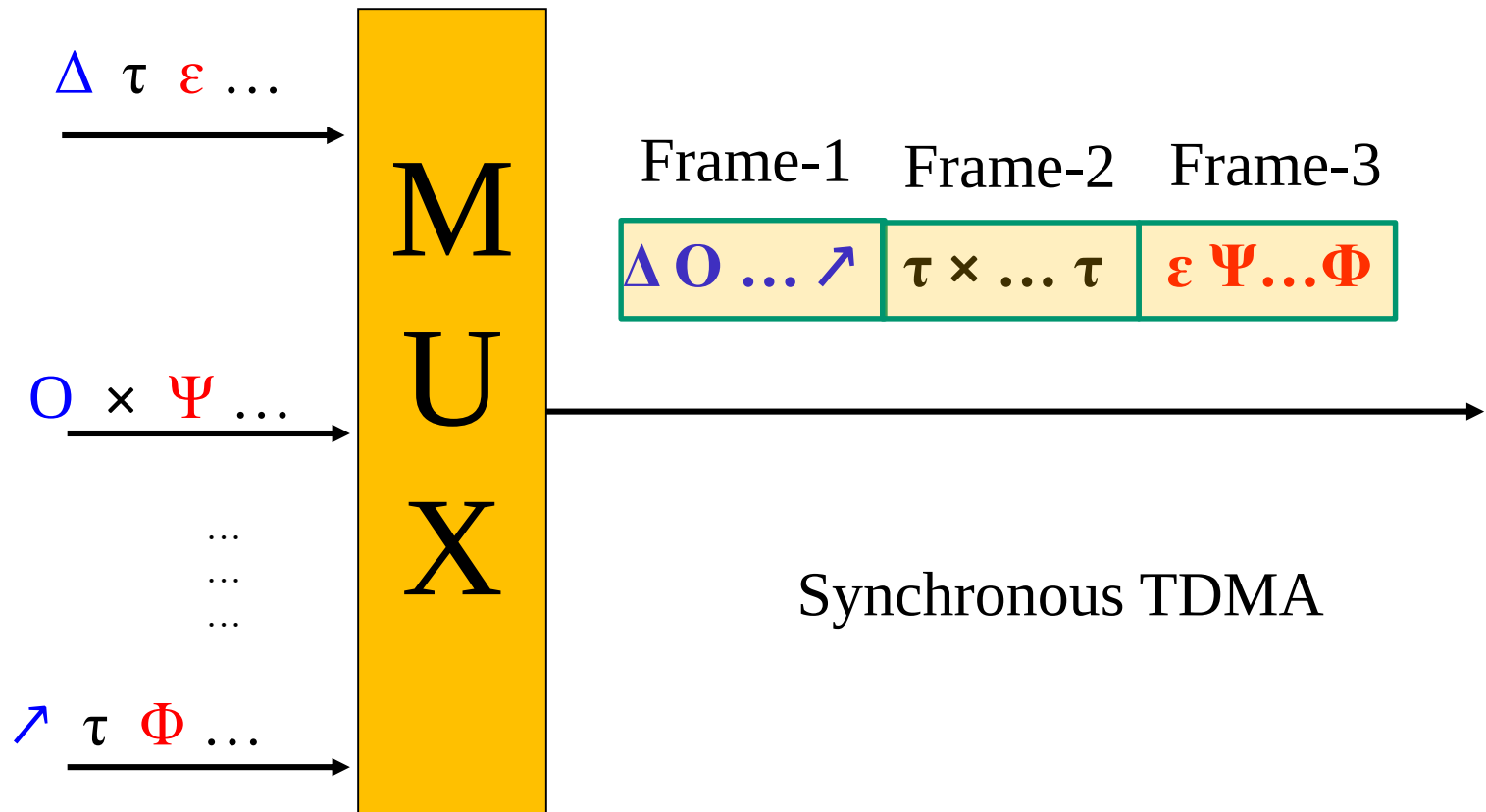


Asynchronous Transfer Mode (ATM)

Definition

- ✓ ATM is **asynchronous** in the sense that cells on TDMA frame from an individual user is not necessarily **periodic** as found in synchronous TDMA.
- ✓ ATM uses statistical (asynchronous) time-division multiplexing i.e. the multiplexer fills a slot with a cell from any input channel that has a cell; the slot is empty if none of the channels has a cell to send shown in next slide.
- ✓ **Transfer mode** is a term intended to signify that it is a multiplexing and switching technique.



ATM Multiplexing

Fig. below shows how cells from three inputs are multiplexed. At the first tick of the clock, channel 2 has no cell (empty input slot), so MUX fills the slot with a cell from third channel. The output of the MUX will be only empty when all input lines are empty together.

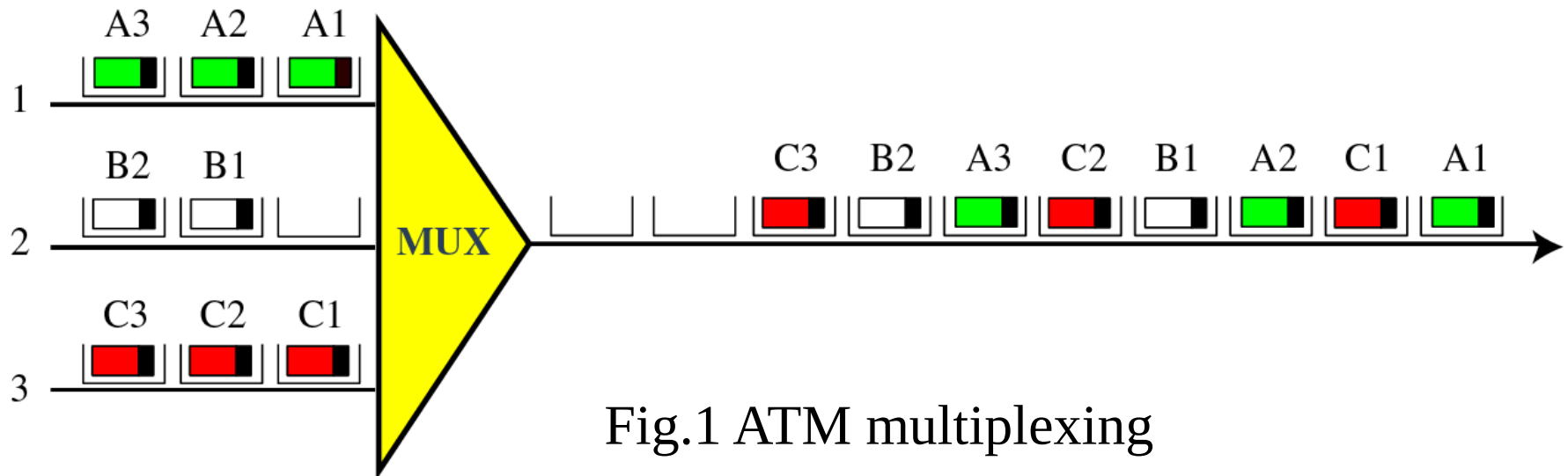
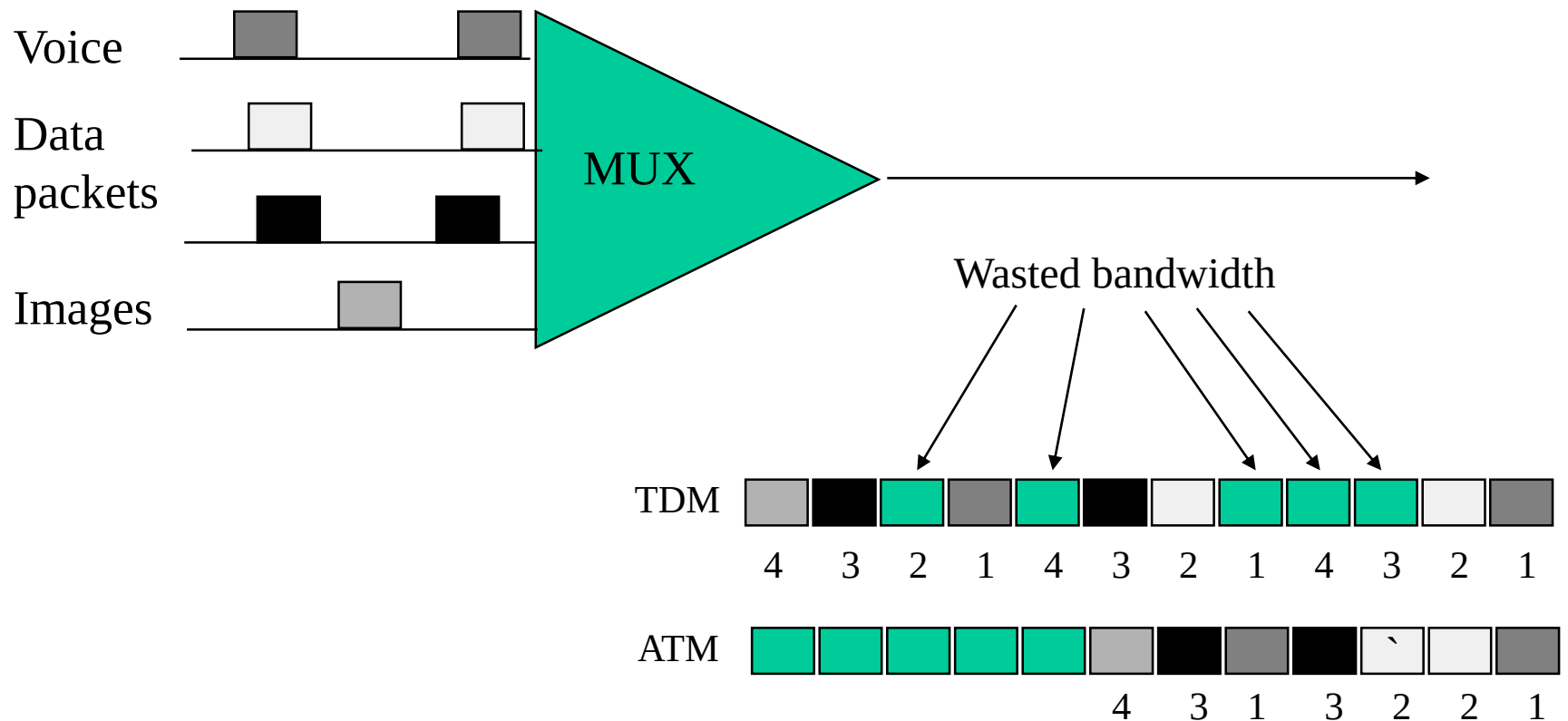


Fig.1 ATM multiplexing

ATM switches or multiplexes data in asynchronous mode.

Asynchronous Transfer Mode (ATM)



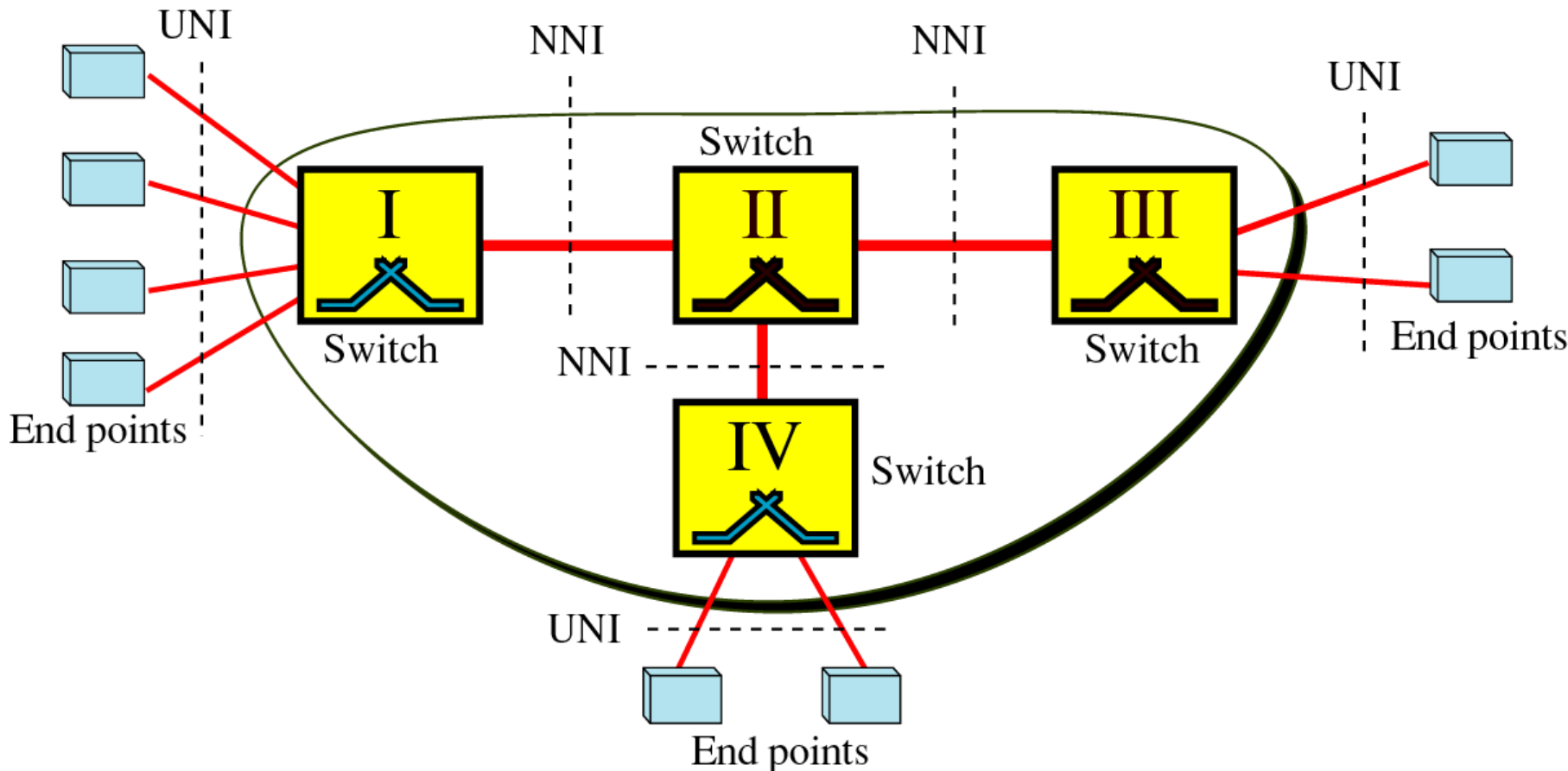
ATM is a packet-oriented (also connection oriented) transfer mode which allows multiple logical connections to be multiplexed over a single physical interface. The information flow on each logical connection is organized into fixed-size packets called cell.

Why small cell?

- ❖ It is easy to build hardware routers to handle short, fixed-length cells. Variable length IP packets have to be routed by software, which is slower process.
- ❖ Hardware can copy one incoming cell to multiple output lines easily which is suitable for TV broadcasting.
- ❖ Small cell can ensure QoS during network congestion.

Architecture of an ATM Network

The user access devices, called the end points are connected through a user-to-network interface (UNI) to the switches inside the network. The switches are connected through network-to-network interface (NNI)

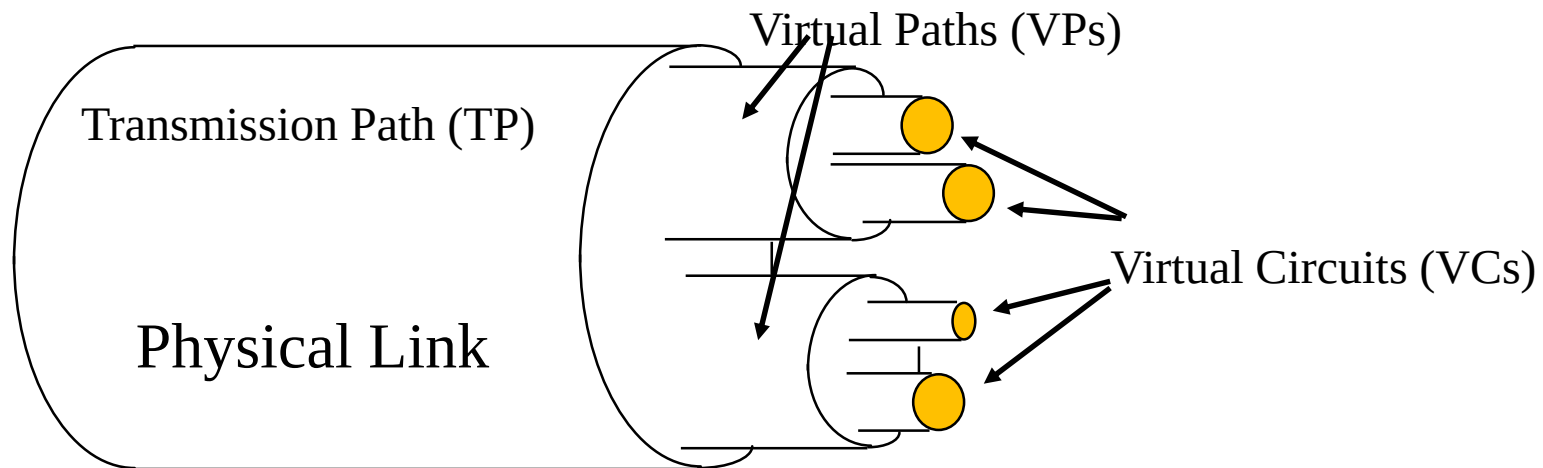


Virtual Connection

There exist two types of ATM logical connections:

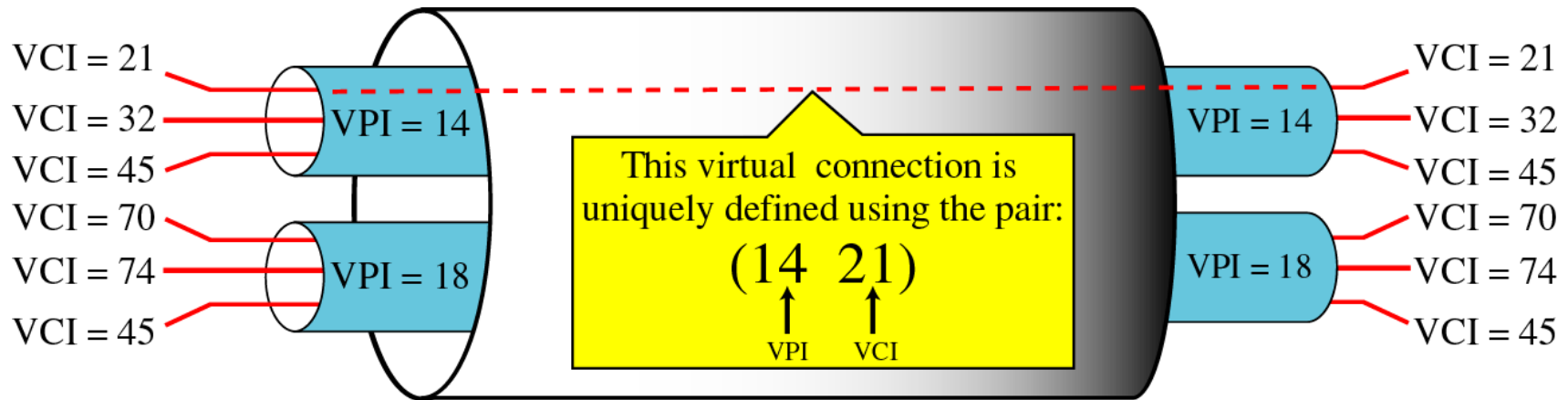
- Virtual Path Connections (VPs)
- Virtual Channel Connections (VCs).

A **transmission path (TP)** is the physical connection (wire, cable, satellite etc.) between an end point and switch or between two switches. A transmission path can be considered as the set of all highways that directly connect two cities. A transmission path is divided into several **virtual paths (VPs)**. Think virtual path as a highway between two cities. Each lane of the highway (virtual path) is considered as **virtual circuit (VCs)**.



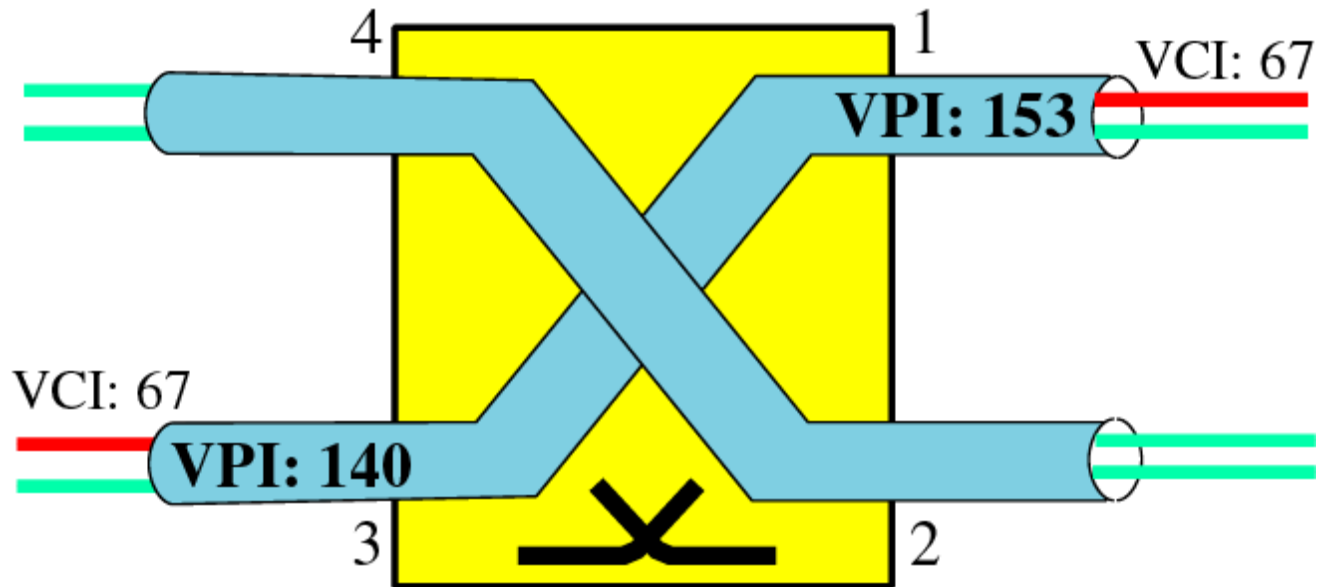
Identifiers

Each connection is identified by a Virtual Path Identifier (VPI)/Virtual Circuit Identifier (VCI) combination, which can be found in the header of all ATM cells carrying information about that specific connection. A group of VC links, identified by a common value of VPI.



ATM Switching

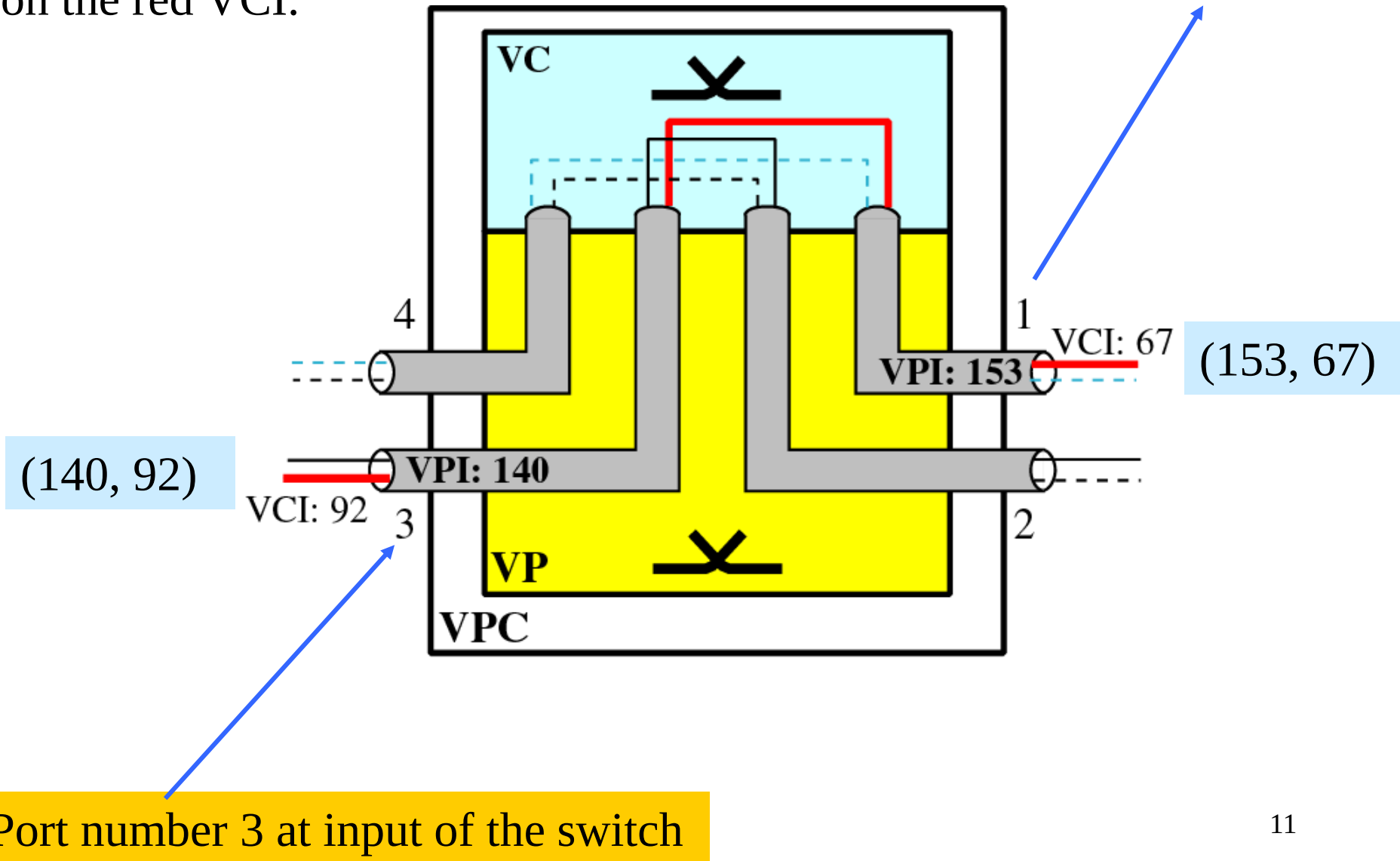
A Conceptual View of a VP Switch

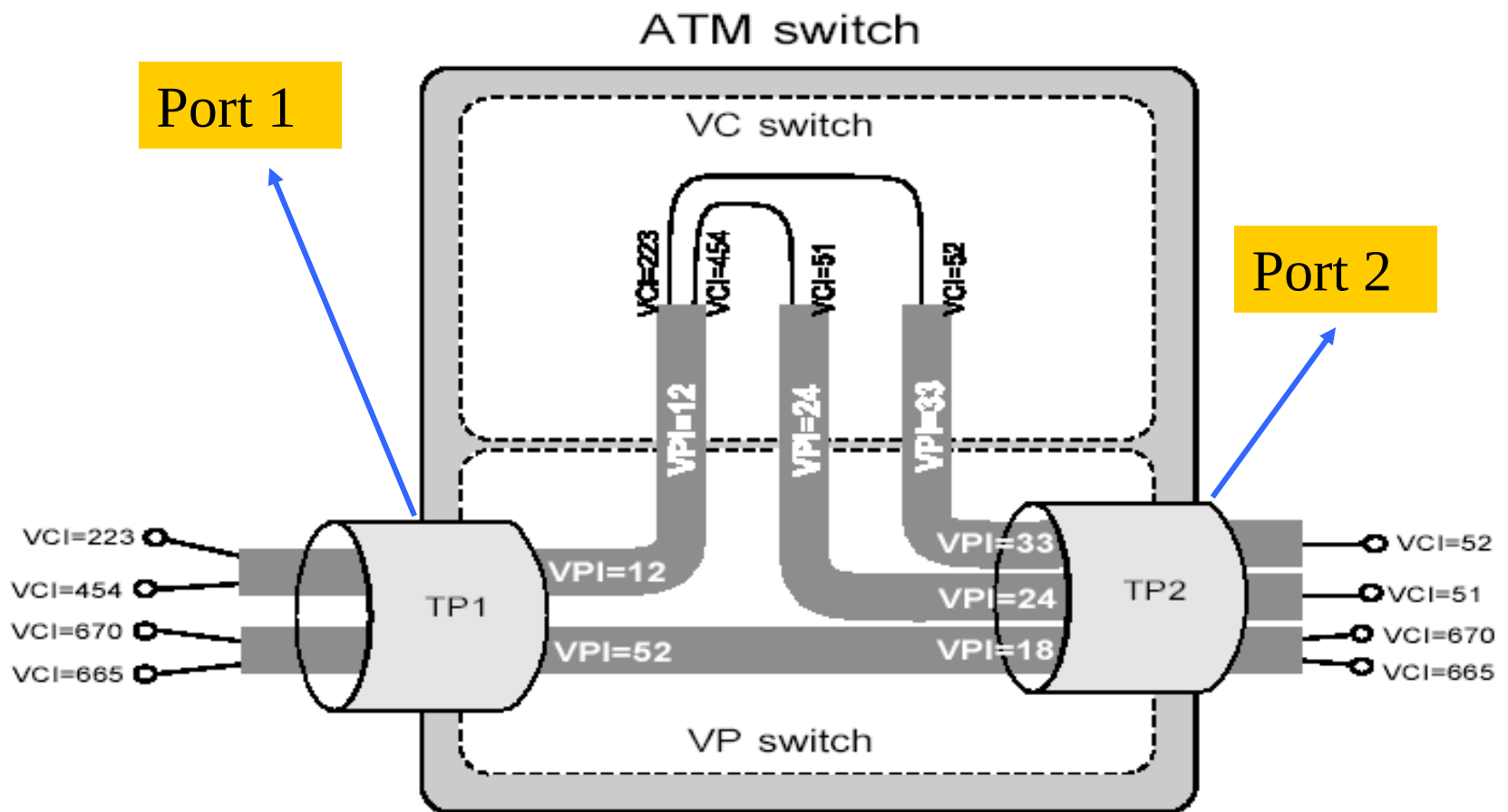


A Conceptual View of a VP and VC Switch

Let us concentrate on the red VCI.

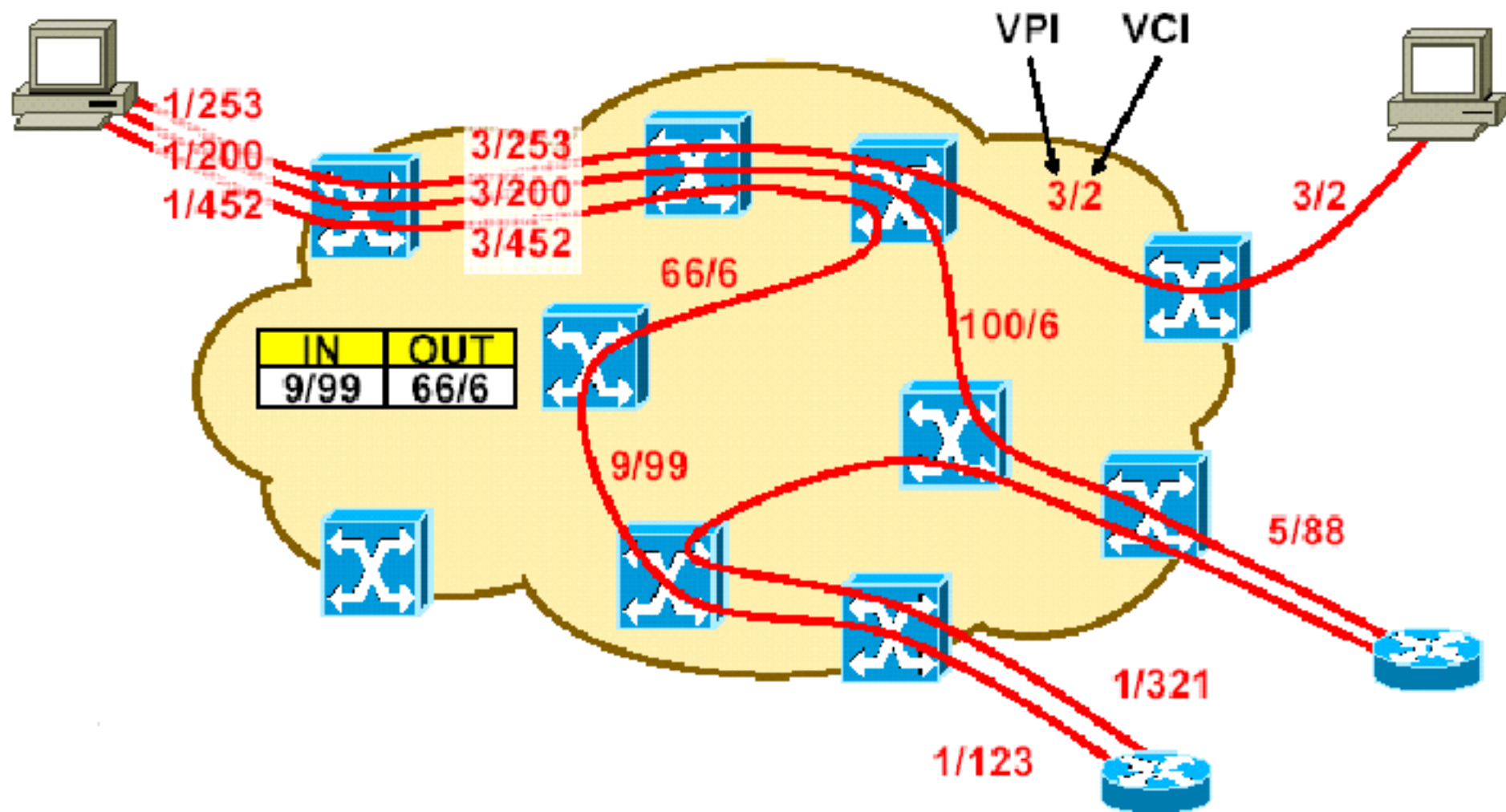
Port number 1 at output of the switch

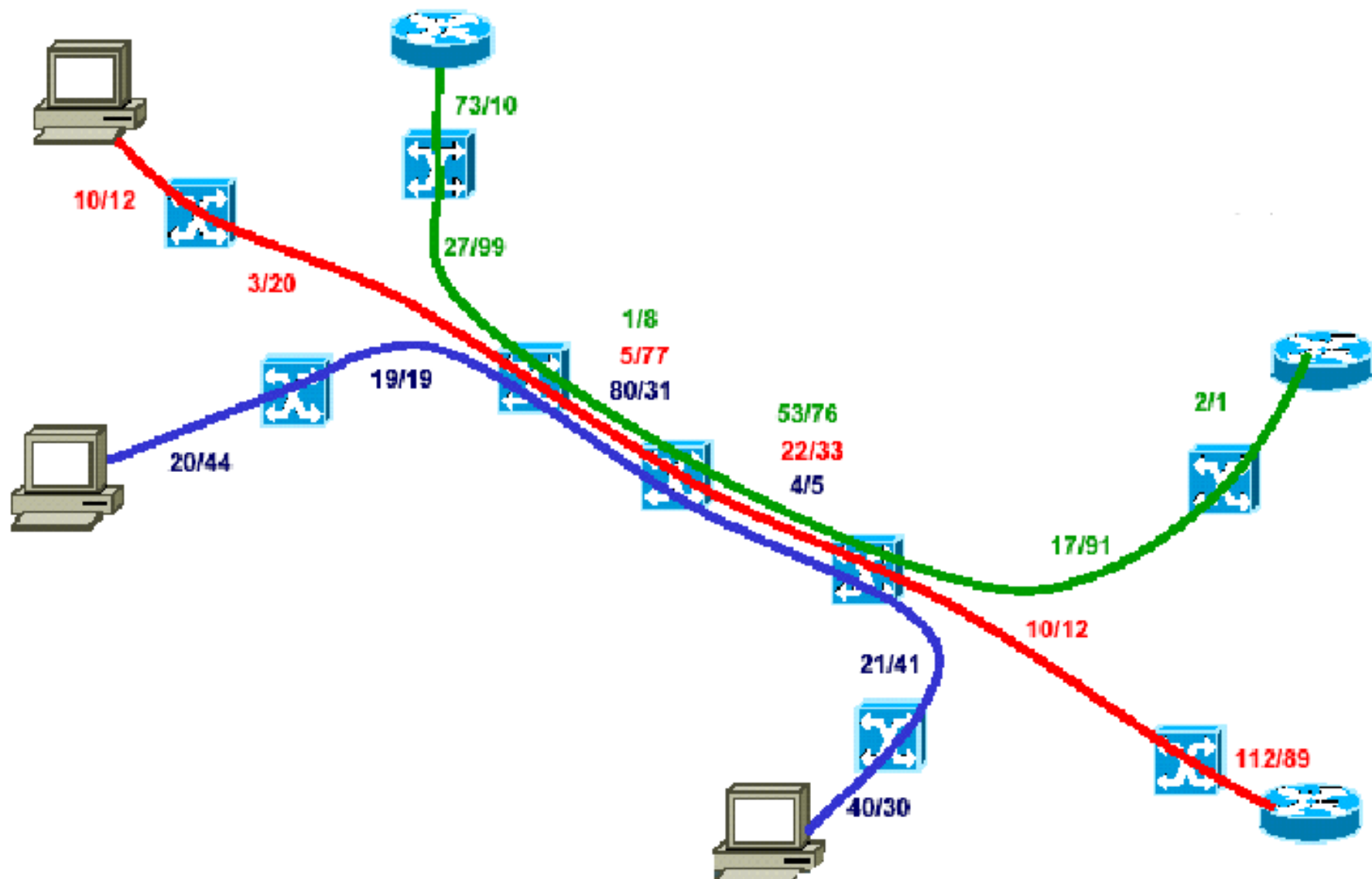




Routing Table

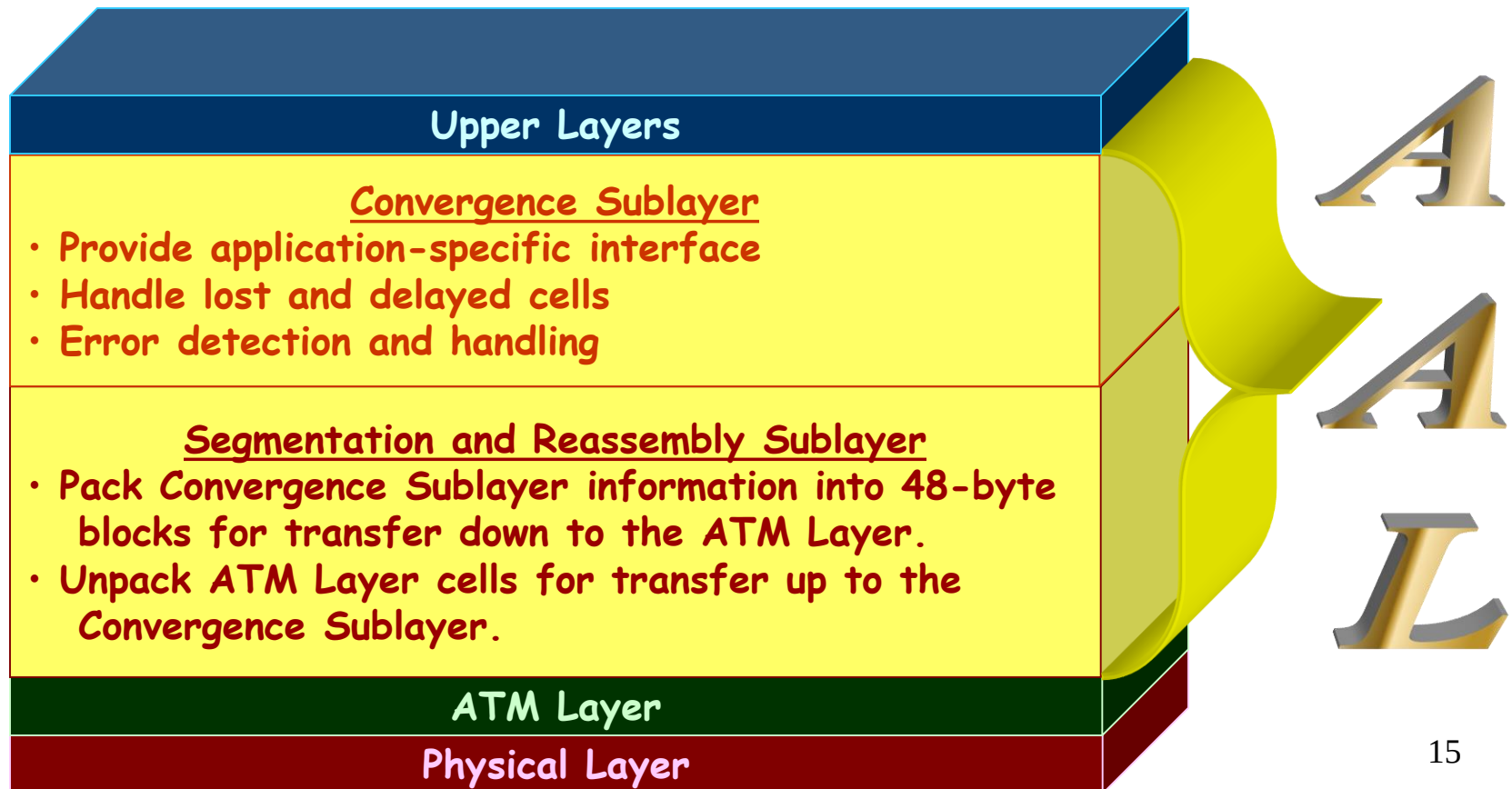
Input			Output		
Port	VPI	VCI	Port	VPI	VCI
1	12	223	2	33	52
1	12	454	2	24	51
1	52	670	2	18	670
1	52	665	2	18	665

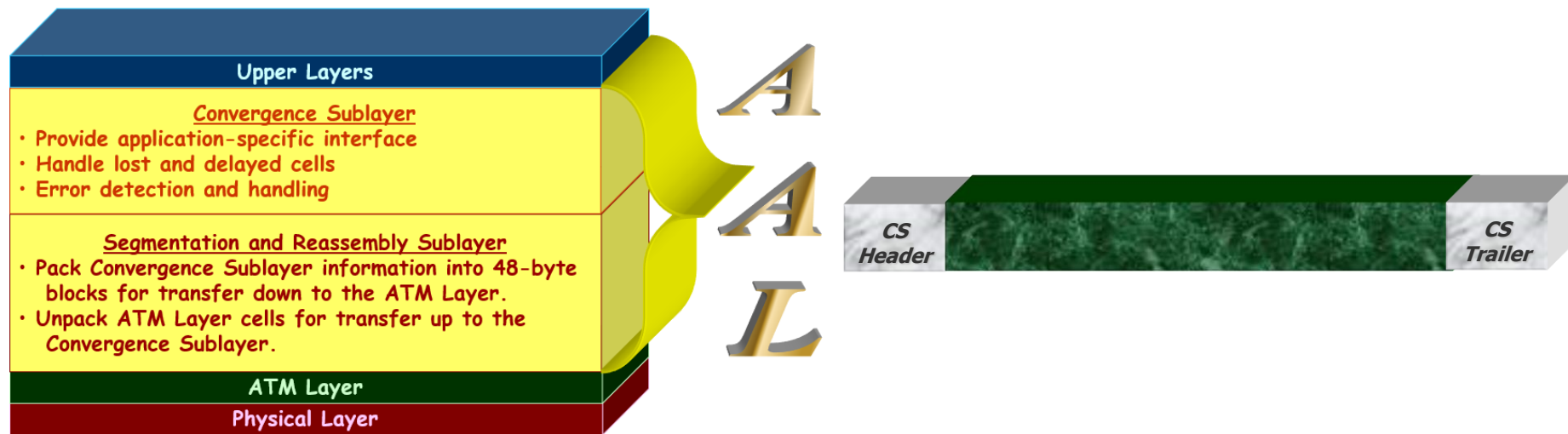




ATM Layered Architecture

ATM has its own reference model different from OSI or TCP/IP model. The ATM standard defines three layers. They are **application adaptation layer (AAL)**, the **ATM layer** and **physical layer**.



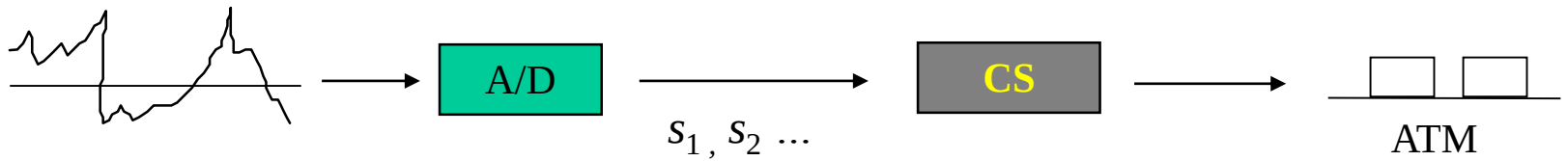


Application Adaption Layer

- ✓ The AAL layer is organized into logical sublayers: The **convergence sub-layer** and **segmentation and reassembly sub-layer**.
- ✓ The **convergence sub-layer** makes it possible to have ATM systems offer different kinds of services of different applications (like file transfer, video on demand etc.). It adds CS head and tail on message segment.

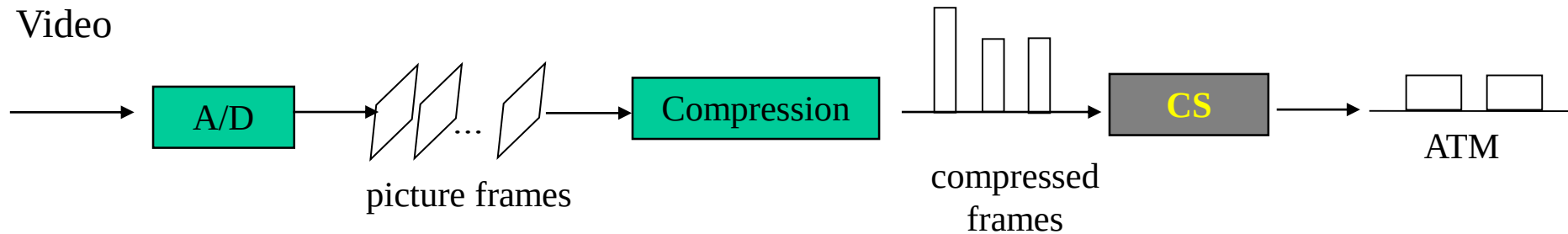
Convergence sub-layer

Voice

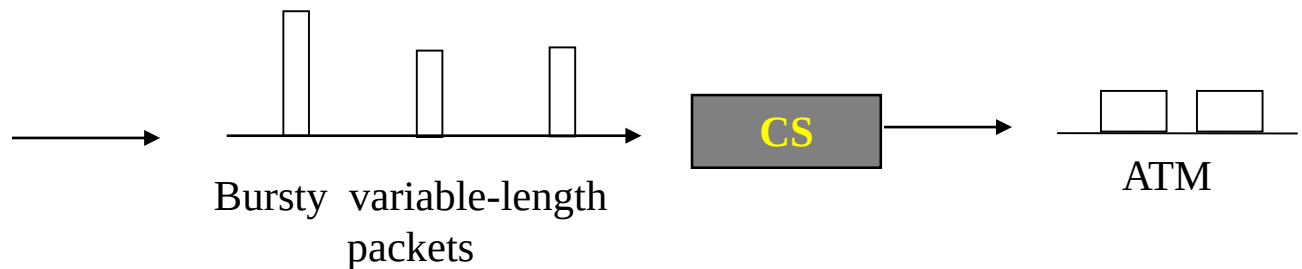


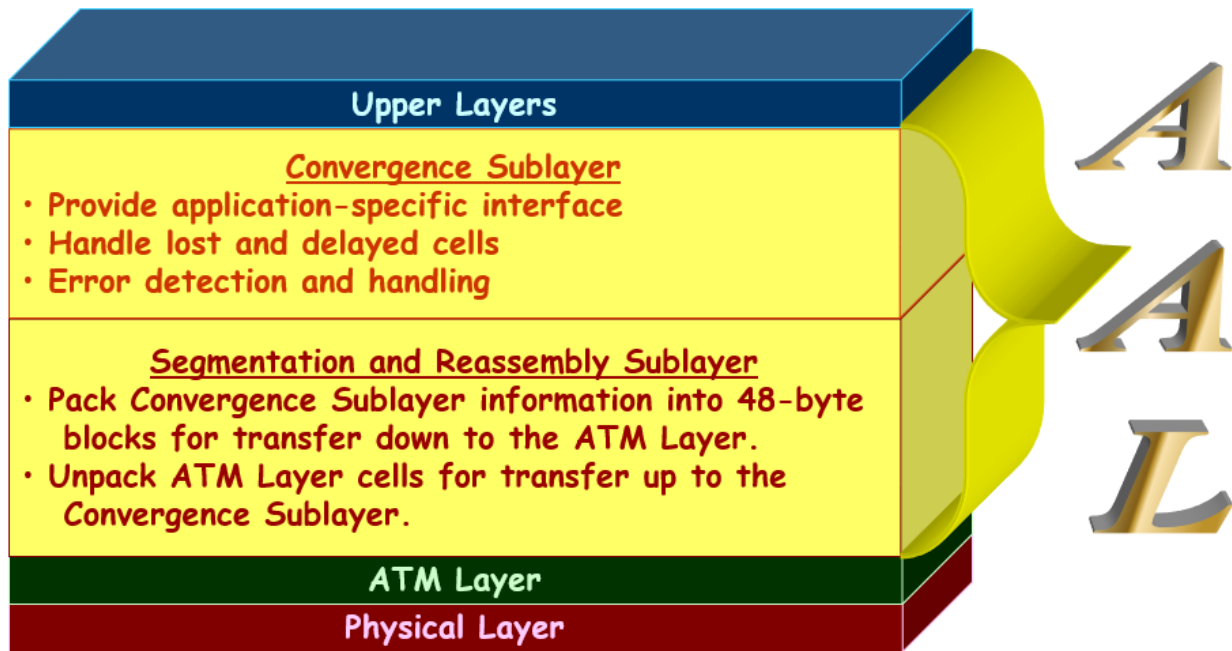
Digital voice in PCM

Video



Data





The **segmentation and reassembly** sublayer is responsible to break up the message (from CS sublayer) into cells on transmission side and puts them back together again at the destination side.

Convergence Sublayer



Segmentation and Reassembly Sublayer

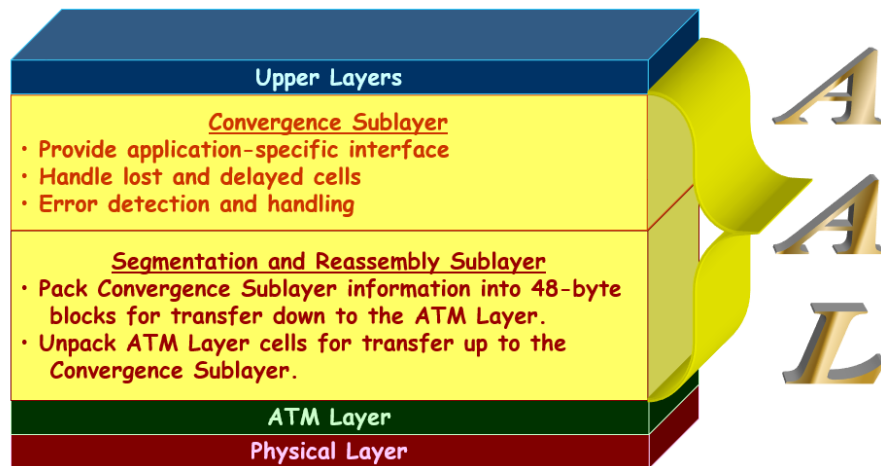


ATM layer

It accepts 48byte segments from AAL and transforming them into 53 bytes cells by addition of 5-byte header i.e., full ATM cell. The ATM layer provides routing, traffic management, switching and multiplexing services.

Physical layer

The physical layer is responsible for transmission of ATM cells as a bit-stream across the physical medium like Ethernet or wireless LAN.



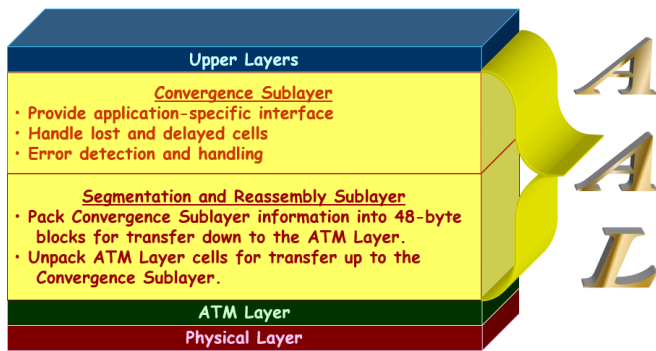
Segmentation and Reassembly Sublayer



ATM Layer



Function of layers on transmitting side



A
A
L

Application Layer



Convergence Sublayer



Segmentation and Reassembly Sublayer



Segmentation and Reassembly Sublayer (continued)



ATM Layer

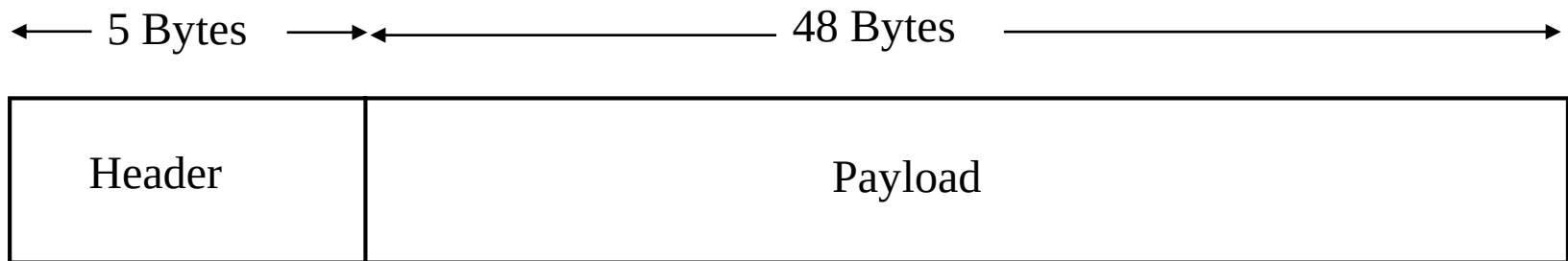


ATM Cell Format

ATM Cells

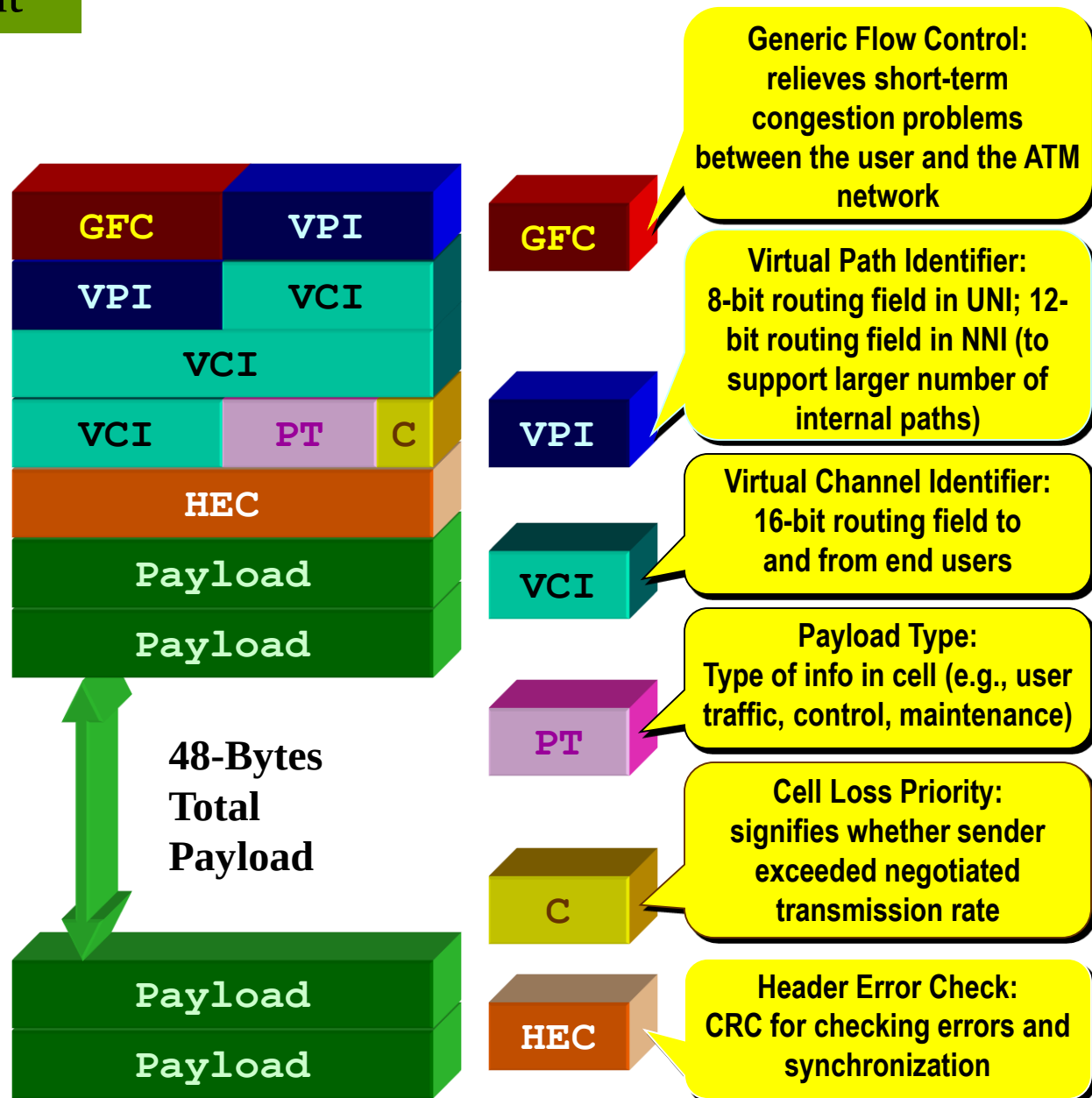
The asynchronous transfer mode makes use of fixed-size cells, consisting of 5-octat header and 48-octet information field

Basic ATM Cell Format

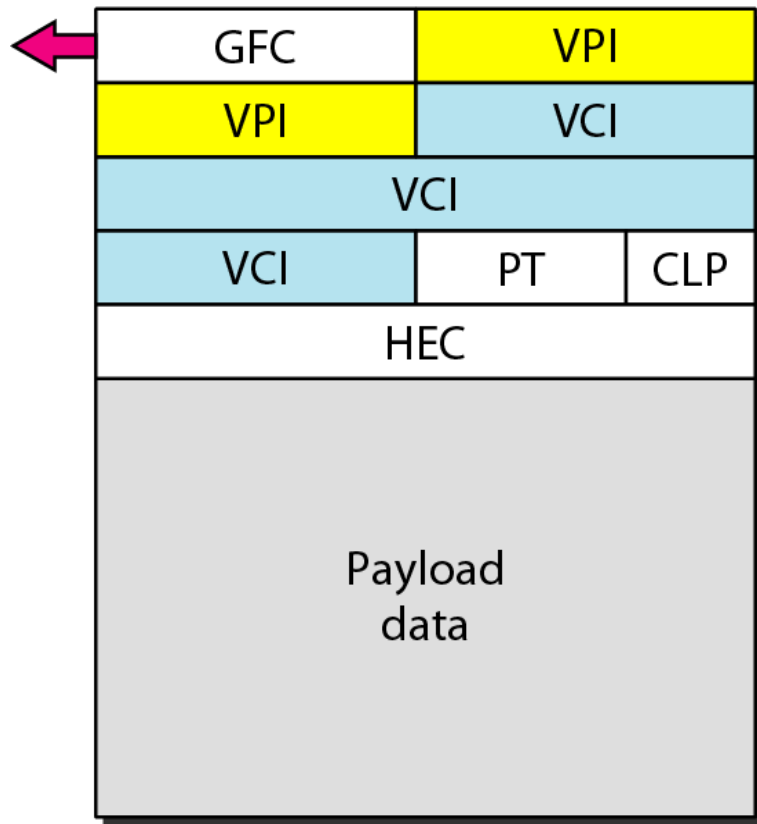


- ❑ Use of small cells reduce queuing delay for high priority cell.
- ❑ Fixed-size cells can be switched more efficiently.

ATM Cell Format

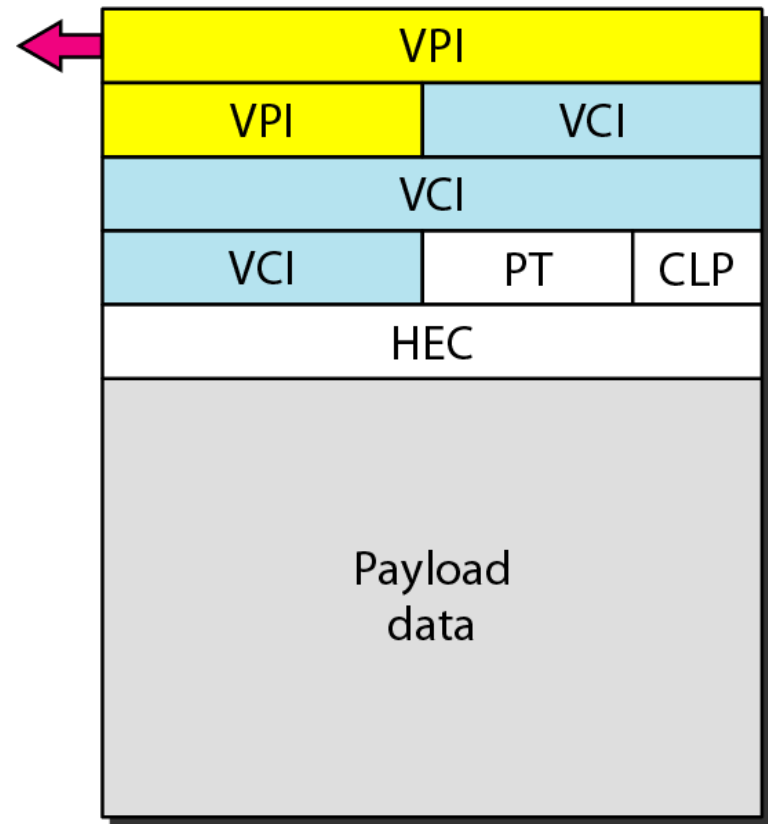


GFC: Generic flow control
VPI: Virtual path identifier
VCI: Virtual circuit identifier



UNI cell

PT: Payload type
CLP: Cell loss priority
HEC: Header error control



NNI cell

GFC 4bits	VPI 8bits	VCI 16bits	PT 3bits	CLP 1bits	HEC 8bits	Payload 48bytes
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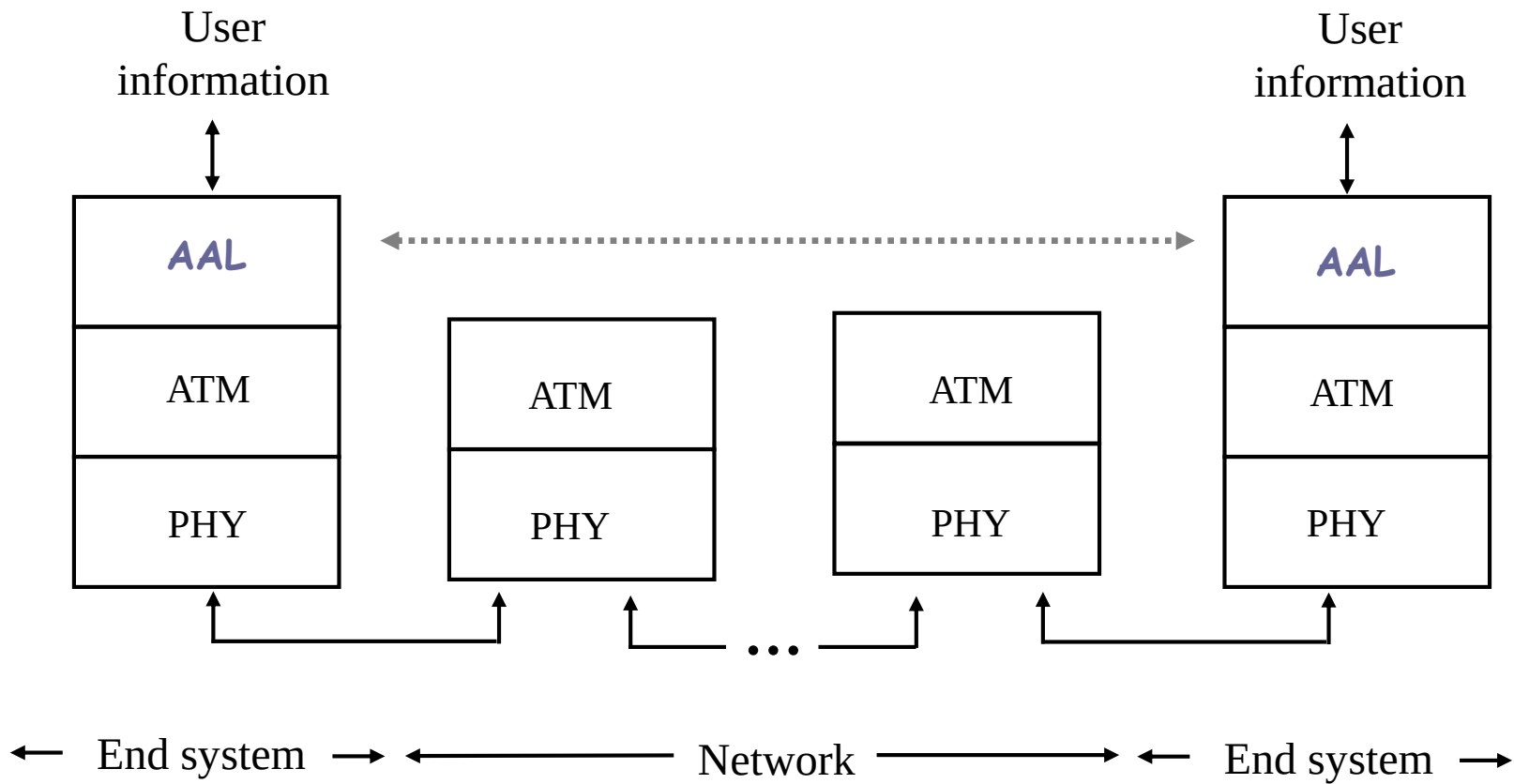
UNI Cell

VPI 12bits	VCI 16bits	PT 3bits	CLP 1bits	HEC 8bits	Payload 48bytes
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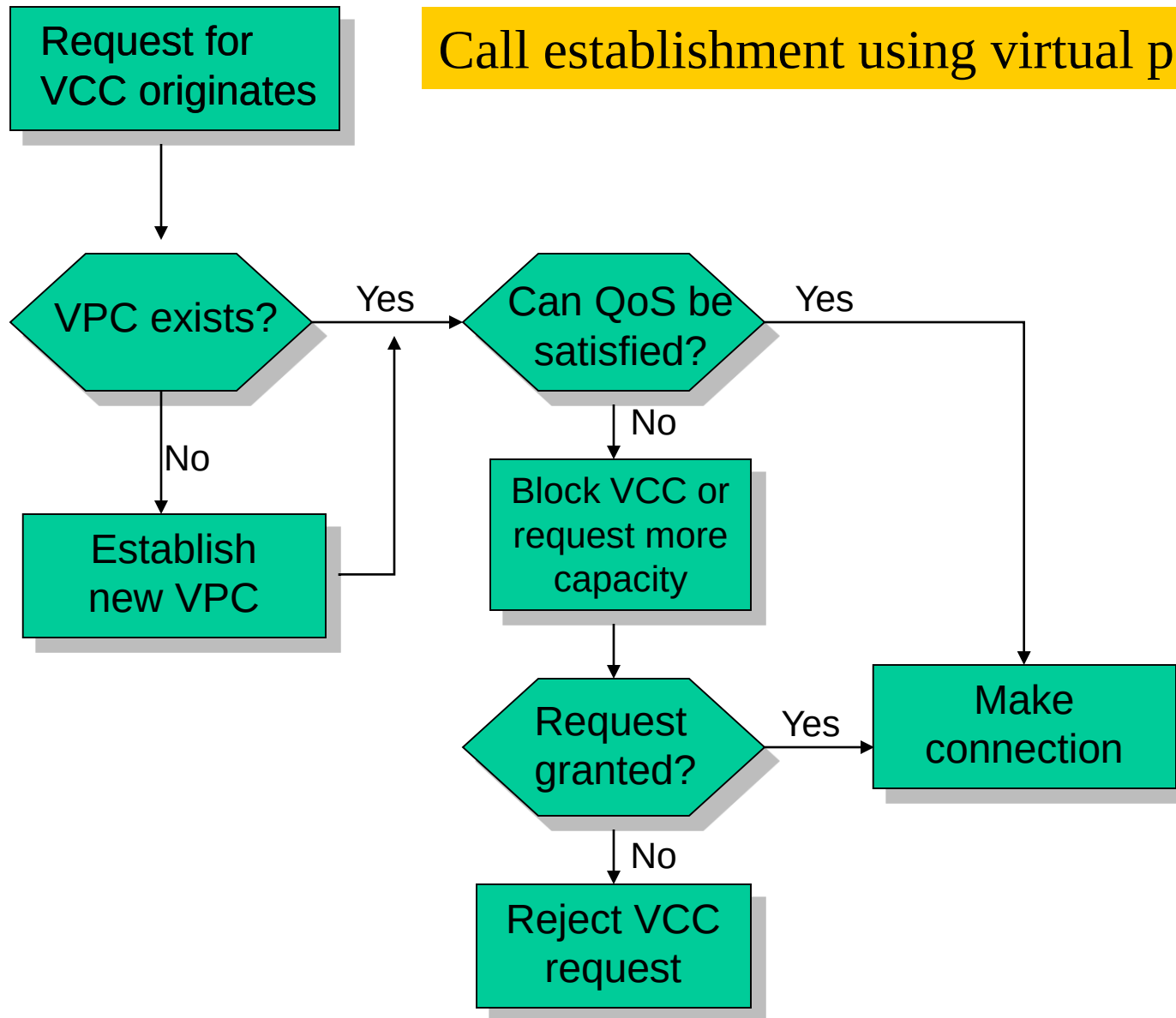
NNI Cell

The CLP bit allows a two level loss priority (CLP = 0 for high, CLP = 1 for low) be specified explicitly for individual cells. Low loss priority cells should be discarded before cells of high loss priority if congestion occurs and necessitates a loss of cells.

HEC uses CRC code



Call establishment using virtual path



Modeling of M/D/1 asynchronous traffic (ATM system)

Most of the traffics in telecommunications network follow exponential arrival and exponential service time distribution like $M/M/n/K/N$ as complete notation. In ATM network service time of each cell/packet is fixed hence deterministic service time traffic like $M/D/n/K/N$ is applicable to detect traffic parameters for that network. In this section a traffic model of single server case, where arrival of traffic is Markovian but the service time is deterministic i.e. $M/D/1$ traffic.

x cells in the queue of router

Let,

$P_t(x) = P\{\text{The system in state } x \text{ at time } t\}$

$a_i = P\{\text{during time interval } (t, t+t_m) \text{ } i \text{ cells arrive}\}$

$t_m = \text{Cell duration} = 1 \text{ time unit}$

$A = \lambda t_m$; is the mean of Poisson's offered traffic

To reach the state $x(t+1)$ from $x(t)$ transitions take place shown in table-1.

Table-1

State $x(t)$	Events (s)	$P\{\text{events}(s)\}$	State $x(t+1)$
0	No arrival	a_0	0
0	1 arrival	a_1	1
0	$x>1$ arrival	$a_{x>1}$	$x>1$
1	No arrival	a_0	0
1	1 arrival	a_1	1
1	2 arrivals	a_2	2
1	$x>2$ arrivals	$a_{x>1}$	$x>2$
2	No arrival	a_0	1
2	1 arrival	a_1	2
2	2 arrivals	a_2	3
2	$x>2$ arrivals	$a_{x>1}$	$1+x>2$
3	No arrival	a_0	2
3	1 arrival	a_1	3
3	2 arrivals	a_2	4
3	$x>2$ arrivals	$a_{x>1}$	$2+x>2$
x	No arrival	a_0	$x-1$
x	1 arrival	a_1	x
x	2 arrivals	a_2	$x+1$
x	$x>2$ arrivals	$a_{x>1}$	$x-1+x>2$

Initially 0 cells
in the queue

a_i means i cells
arrival within a
cell duration t_m

Initially 3 cells
in the queue

State $x(t)$	Events (s)	$P\{\text{events}(s)\}$	State $x(t+1)$
0	No arrival	a_0	0
0	1 arrival	a_1	1
0	$x > 1$ arrival	$a_{x>1}$	$x > 1$
1	No arrival	a_0	0
1	1 arrival	a_1	1
1	2 arrivals	a_2	2
1	$x > 2$ arrivals	$a_{x>1}$	$x > 2$
2	No arrival	a_0	1
2	1 arrival	a_1	2
2	2 arrivals	a_2	3
2	$x > 2$ arrivals	$a_{x>1}$	$1+x > 2$
3	No arrival	a_0	2
3	1 arrival	a_1	3
3	2 arrivals	a_2	4
3	$x > 2$ arrivals	$a_{x>1}$	$2+x > 2$
x	No arrival	a_0	$x-1$
x	1 arrival	a_1	x
x	2 arrivals	a_2	$x+1$
x	$x > 2$ arrivals	$a_{x>1}$	$x-1+x > 2$

Balanced equations from above table could be detected as,

$$P_{t+1}(0) = P_t(0)a_0 + P_t(1)a_0$$

$$P_{t+1}(1) = P_t(0)a_1 + P_t(1)a_1 + P_t(2)a_0$$

$$P_{t+1}(2) = P_t(1)a_2 + P_t(2)a_1 + P_t(3)a_0$$

etc.

Balanced equations from above table could be detected as,

$$P_{t+1}(0) = P_t(0)a_0 + P_t(1)a_0$$

$$P_{t+1}(1) = P_t(0)a_1 + P_t(1)a_1 + P_t(2)a_0$$

$$P_{t+1}(2) = P_t(1)a_2 + P_t(2)a_1 + P_t(3)a_0 \text{ etc.}$$

In statistical equilibrium state actual point (t) on time axis does not have any influence on the probabilities, become time-independent steady state probabilities $P(x) = P_x$

$$P_0 = P_0a_0 + P_1a_0$$

$$P_1 = P_0a_1 + P_1a_1 + P_2a_0$$

$$P_2 = P_1a_2 + P_2a_1 + P_3a_0 \text{ etc.}$$

These equations could be solved recursively say,

$$P_1 = \frac{P_0(1 - a_0)}{a_0}$$

$$P_2 = \frac{-P_0 a_1 + P_1(1 - a_1)}{a_0}$$

Here a_i follows Poisson's process,

$$a_i = \frac{(\lambda t_m)^i}{(\lambda t_m)!} e^{-\lambda t_m}$$

Basic equation of carried traffic,

$$\bar{X} = \sum_{x=0}^{\infty} x P_x = 0.P_0 + 1.P_1 + 2.P_2 + 3.P_3 + \cdots \quad \dots \quad \dots$$

Number of served packets

Probability of x packets in queue

In M/D/1 any number packets x may stay in queue but only one packet will be served at a time. Therefore, equation of carried traffic will be,

$$\bar{X} = \sum_{x=1}^{\infty} 1.P_x = 1 - P_0 = A$$

Since the size of the buffer is infinity therefore carried traffic and offered traffic will be same i.e. no loss of traffic.

Since buffer is infinite hence carries traffic,

$$\sum_{x=1}^{\infty} P_x = 1 - P_0 = A$$

$$\therefore P_0 = 1 - A$$

$$P_1 = \frac{P_0(1 - e^{-A})}{e^{-A}} = P_0(e^A - 1)$$

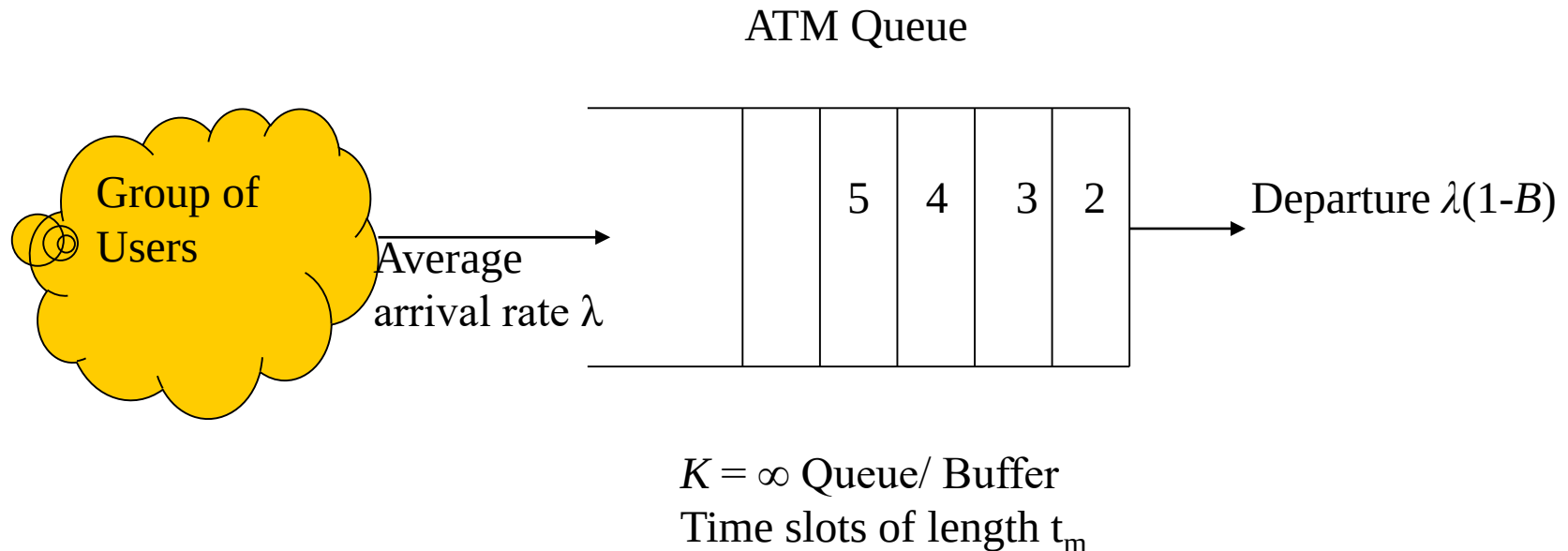
$$P_2 = e^{-A}[-P_0 A e^{-A} + P_1(1 - A e^{-A})] = (1 - A)e^{-A}[e^A - 1 - A]$$

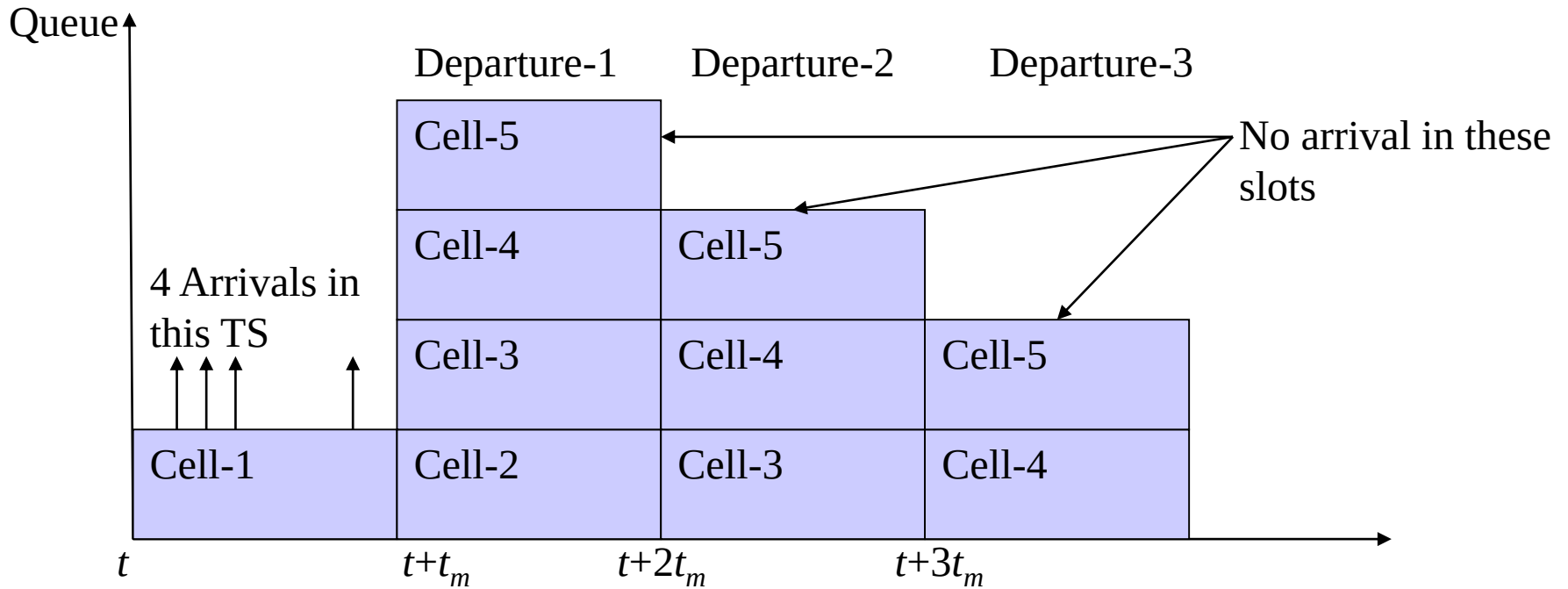
...

...

$$P_x = (1 - A) \sum_{k=1}^x (-1)^{x-k} e^{kA} \left[\frac{(kA)^{x-k}}{(x-k)!} + \frac{(kA)^{x-k-1}}{(x-k-1)!} \right]; x \geq 2$$

The asynchronous M/D/1 formulas are well valid for a synchronous M/D/1 queue with internal server (S-IS M/D/1). In this system during the time slot (cell time in ATM system) cell arrive according to Poisson process, which is defined by a single parameters, the mean arrival rate λ .





At the end of the TS the queue (internal server) first removes a cell (if any) from the queue (its service time is complete) and then stores the new cells. Each cell spends in the queue a minimum time of 1 slots (cell service time).

During the interval $[0, t_m]$, 4 cells arrives. The probability state at $t+t_m$ is P_4 . The waiting time of cells will be $t_w = 4t_m$.

Therefore, $P(t_w = 4t_m) = P_4$. In generalized form, $P(t_w = kt_m) = P_k$.

For ‘integer’ waiting times that are integer multiples of service time the waiting time distribution is equal to the system state distribution. Let $P\{t_w \leq k.t_m\}$ be the probability that waiting time is less or equal than k service times and P_x is state probability, then,

$$P\{t_w \leq k.t_m\} = P_0 + P_1 + P_2 + \dots \dots \dots + P_k$$

$$P\{t_w \leq k\} = \sum_{i=0}^k P_i$$